A study related to Air India Data Breach Incident 2021

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Data Governance and Ethics

Introduction

In a century where technology has a significant impact on people's daily lives, cyber security concerns are on the rise. One of the most frequent situations that occur today is a data breach. A data breach incident occurs when a third party improperly acquires access to personal and private data using a variety of methods. Nowadays, "big data" data breaches are more prevalent due to the wide range of internet platforms that individuals use. In this essay, we talk about the occurrence involving the Air India data breach that occurred in 2021.

About the Incident:

Air India Data Breach Incident 2021

In May 2021, it was revealed that the Indian airline "Air India" had been the victim of a cyberattack that exposed the personal information of roughly 4.5 million individuals worldwide. Information such as a passport number, credit card information, a person's name and birthdate, and ticket data were among the details exposed. According to reports, the information that was exposed dates from 26 August 2011 until 20 February 2021. The airlines declared that the reason for the data leak was due to the data breach from Air India's data processor, "SITA," a technology-based business with headquarters in Switzerland. The organisation has clients from almost 200 different nations and specialises in information technology and air transport communication. The data breach occurred in the latter week of February, Air India learned about it in March, and the public learned about it in May 2021.

Damages that can cause due to Big Data Breach:

The loss of data, especially big data, which is typically generated by sizable organisations, can have significant detrimental effects. According to the International Data Corporation (IDC), data breaches will affect a quarter of the world's population by 2020. Which, in a sense, came to pass, and the number of events kept rising. In addition to having an adverse effect on the individuals because of the personal information that should be kept private, it can also have an adverse effect on the organisations.

Some of the major impacts are:

Financial loss: Depending on the situation, this may occur. Organizations experience financial loss as a result of the effects of data breaches, including loss from having to compensate customers, loss in share value, and increased security costs after trying to contain the breach.

Damage to reputation: In today's environment, news spreads quickly, and as a result, a data breach occurrence is revealed within minutes. The organization's reputational worth may suffer as a result. After the occurrence, the company may suffer a significant loss, and the loss of customers' faith may lower the organization's value.

Operational disruption: A data breach may result in the loss of data from a specific time period or possibly the full loss of vital data, which will seriously affect corporate operations because there won't be any data to work with. And the company can only be restarted if the lost data has been recovered.

Legal concerns: Lawsuits are frequently brought when a data breach involves private consumer information. Recent occurrences simply serve as more evidence that large organisations frequently must pay substantial sums of money to resolve legal claims made against them following security breach incidents.

The brand worth of a large airline that operates not only within India but also around the globe was certain to suffer some harm as the information that was leaked included both the passport information and the credit card information of the individuals. Customers were extremely outraged by the occurrence since this type of information can be utilised in crimes like identity theft and credit card fraud. The fury was compounded by the fact that the incident was brought to light in May even though it occurred in February.

Even though the airlines later stated that the data breach incident had not hurt any particular people and that no incidents had occurred as a result of people's personal information being leaked, the damage to the airlines' reputation was still done, and even though the financial losses were not made public, it was still known that the airlines had suffered losses as a result of the data breach.

The government opted to sell the airlines due to the financial loss and earlier obligations, which totaled over $9 billion. Later in September, the business "TATA Sons Ltd" purchased "Air India" from the government.

What did Air India do about the security breach?

Following the data breach incident, the airlines announced that they will begin a legal inquiry into the matter and discussed taking action about the compromised server. They also suggested that clients whose personal information had been compromised reset the passwords and pins associated with their credit card information.

How does the incident affect individuals?

Individuals are unaffected by the data leak as long as their personal information is excluded. However, it could result in some serious issues for the people whose personal information was exposed. Despite the airlines' assurances that the data leak was not utilised maliciously, it might have resulted in serious issues including identity theft, credit card fraud, and cyberattacks.

Political aspects of data:

Data is one of the most valuable resources. Since data is knowledge it can be used in many ways. As the coin has two sides data can be either used for the good and for the bad. Data can either do good to the society or destroy the balance it keeps.

The controversy over the airline data leak could have a negative political influence on the current ruling party and potentially help the opposition. Since "Air India" was a government-owned airline, it might have an impact on the political party in power at the moment. Details of a person's passport and credit card, which are private information, are among the leaked data. These types of data leaks from a government-affiliated organisation seriously damage the government's reputation. Additionally, by highlighting this weakness in the upcoming election, the opposition party may win

support. Realizing the risk of damage in reputation may also be the reason for the government to sell the airlines to a private company.

It should also be considered that a data can also be used to manipulate voting system and a leaked data with personal details are easier to manipulate and even blackmail the individuals who got their data leaked. The availability of data can also be used to manipulate peoples interest as well.

A new generation of data-savvy politicians who can make sense of it is emerging as the way politics is conducted changes. There are two ways that politicians can exploit data analysis, which is the problem. Either they can utilise it for the benefit of everyone or they can use it to further their own interests.

A political campaign can succeed or fail depending on the data, or more particularly, how it is interpreted. However, while it is true that it might assist individuals in winning office, it can also assist them in performing their duties much more successfully and effectively.

But a data breach from a government-run organisation can also harm the standing of officials in the executive and legislative branches of government, including in this case the minister in charge of air travel.

Ethical aspects of data:

Data ethics is the systematic organisation, defence, and advocacy of conceptions of good and wrong behaviour in connection to data, especially personal data. The following factors are covered by data ethics. Ownership, Transparency in transactions, Consent, Confidentiality, Currency, and Openness. Big data ethics is concerned with the data brokers, governments, and major enterprises that gather and disseminate structured or unstructured data.

For want of a better phrase, the episode involving the airlines was unethical. Without their permission, more than 4.5 million people's personal information was made public. Each individual has the right to know how their personal data is being utilised as they are the owners of that information. When an organisation needs a person's personal information, it is also their responsibility to be transparent about how the information will be used. Organizations should always obtain consent from the individual before using their personal information, and the data should be securely safeguarded to prevent leaks.

The ‘Air India’ data breach incident has broken the ethical rule of data. As an organisation maintaining confidential data they should have strongly followed the ethical aspects of the data, since organisations such as these have high responsibility in maintaining the privacy of their customers.
Customers recognise the value of their personal information and value it when it is used responsibly to further their interests and requirements. Even the notion that effective data analytics can reveal hidden motivations and desires excite many people. Such individuals understand how data analytics, when used effectively, can lead to fresh discoveries and experiences that weren't previously or consciously acknowledged. Therefore, it is the responsibility of an organisation to maintain their customers' trust.

What should be done?

Even though "Air India" rebranded to do a legal inquiry and reassure its clients that none of the exposed data was misused, it shouldn't have happened in the first place. As a business that handles sensitive personal information including credit card numbers, passport information, and visa

applications, they are required to maintain a high level of security yet have fallen short in this regard.

As an organisation who should follow the ethical aspects of data strictly they should have heightened their security.

Be sure to notify your clients and prospects whenever you are gathering their personal data, even if it seems obvious. However, simply informing individuals that their data is being gathered is no longer sufficient. You must also be eager to explain in detail how that data will be utilised, preferably in a way that will encourage them to quickly divulge accurate information. Protection and privacy although assuring your clients that their data will remain secret is consoling, it happens far too frequently that any sense of security your clients' and prospects' felt is destroyed by an unanticipated data breach. And it follows naturally that after such an incident, they have a substantially lower level of confidence in your abilities. If a breach does occur, contact all parties that are impacted right away. A declaration of the safety precautions taken to make sure this incident is a one-time occurrence ought to go along with this notification. However, showcasing the security mechanisms in place to shield all exposed data is of utmost importance. A lot of businesses are proactively implementing the Privacy by Design framework and embedding desired ethical principles into the planning and development stage of any platform or solution using data as a result of the increasing awareness of the need for privacy. Instead of attempting to change and introduce data ethics later, this kind of preplanning incorporates data ethics into any solution at the time of conception.

Double factor authentication is also a good measure. If it can be designed in such a way where an external source try to hack ones data an alert will be sent to both organisation as well as concerning individual. This will strengthen the security.

Data breaching also cause financial damage and its concerning

One of the best methods to avoid data breaches is to combat ignorance. It's crucial to teach your staff how to prevent data from being hacked. Data security standards-related procedures can be created and updated consistently. It will be evident from this what your organisation expects in terms of statistics. Additionally, by doing this, you'll be able to remind your staff that you take data seriously and demonstrate to them that they should do the same. Your network is continuously monitored by remote monitoring. Your data may occasionally be maliciously deleted in data breaches. Your data should be regularly backed up so that it can be quickly restored in the event of data loss, server failure, or even a natural disaster. Track the data you store on your machines, and periodically purge what is superfluous. The number of locations where you keep sensitive information should be kept to a minimum, and you should keep track of where each location is. Make sure any anything that might contain sensitive information has been destroyed appropriately before you discard it. Physical acts have the potential to result in data breaches, so it's crucial to protect all data, including physical files. Employees should be well-versed in the websites that potentially endanger company systems, such as file-sharing portals. The same holds true for any mobile devices people use while working. Make sure you take the necessary procedures to prevent a security compromise. You can buy security software and set it up to run automatically continuously. hiring a security specialist to handle this for you or simply getting advice on best practises will help you prevent a data breach within your business.

Conclusion:

The data breach regarding the airlines was due to the lack of security measures taken from the part of the organization. As an organisation that maintains highly confidential details of their passengers, it is important to have a very high security securing these data. As the organisation was under the government it diminished the reputation of the government as well. Numerous businesses have often reported cyberattacks, which pose a severe threat to people's privacy. Cybercrime and criminal cyberattacks have increased as a result of the increased digitalization of the world. Taking sudden security measures as well as preventing from such incidents happening is the duty of the organisation. Data are not a naturally occurring resource; instead, some data were chosen to be collected and stored, while others were ignored. It includes both a political and ethical element. Big data has an impact on how we view culture, citizenship, and democracy. When maintaining data we should always consider its political and ethical aspects. Data is not always biased so when concerning data its should be considered sensitive.

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